

NEPA 3D Knowledge Graph Review System

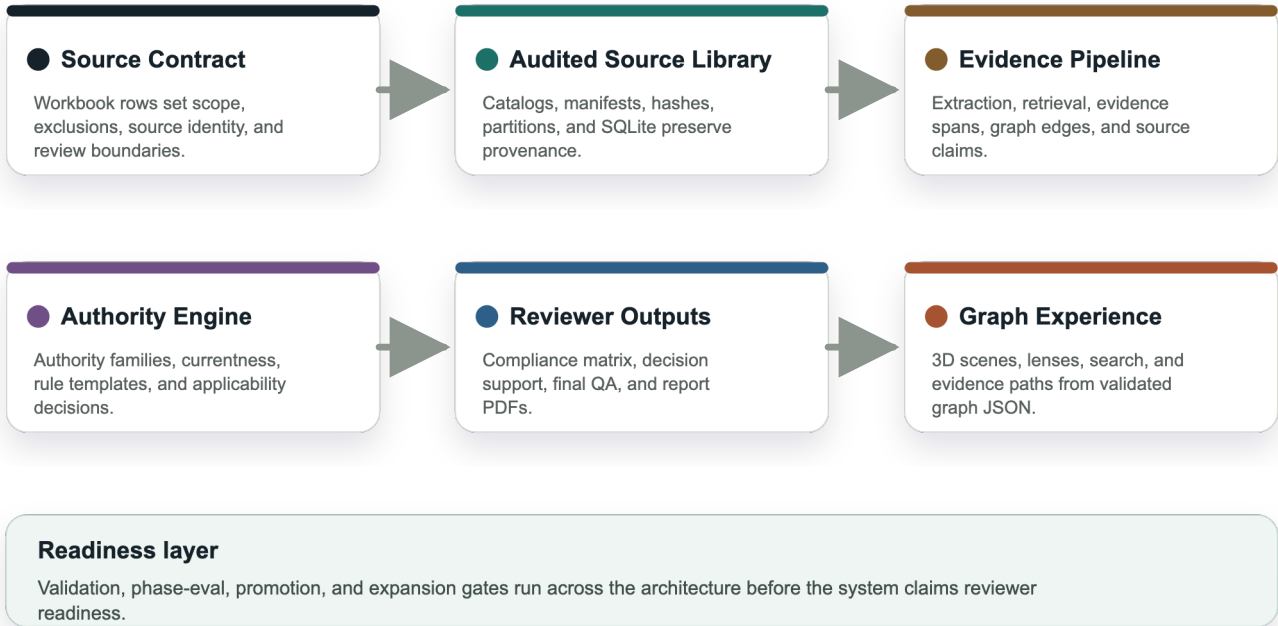
The system turns Region 1 NEPA source material into auditable review intelligence. It preserves source provenance, builds an authority graph, traces evidence to findings, and gates readiness before outputs reach reviewers. The result is a transparent operating layer for compliance review, decision support, and graph exploration.

Current Region 1 implementation snapshot.



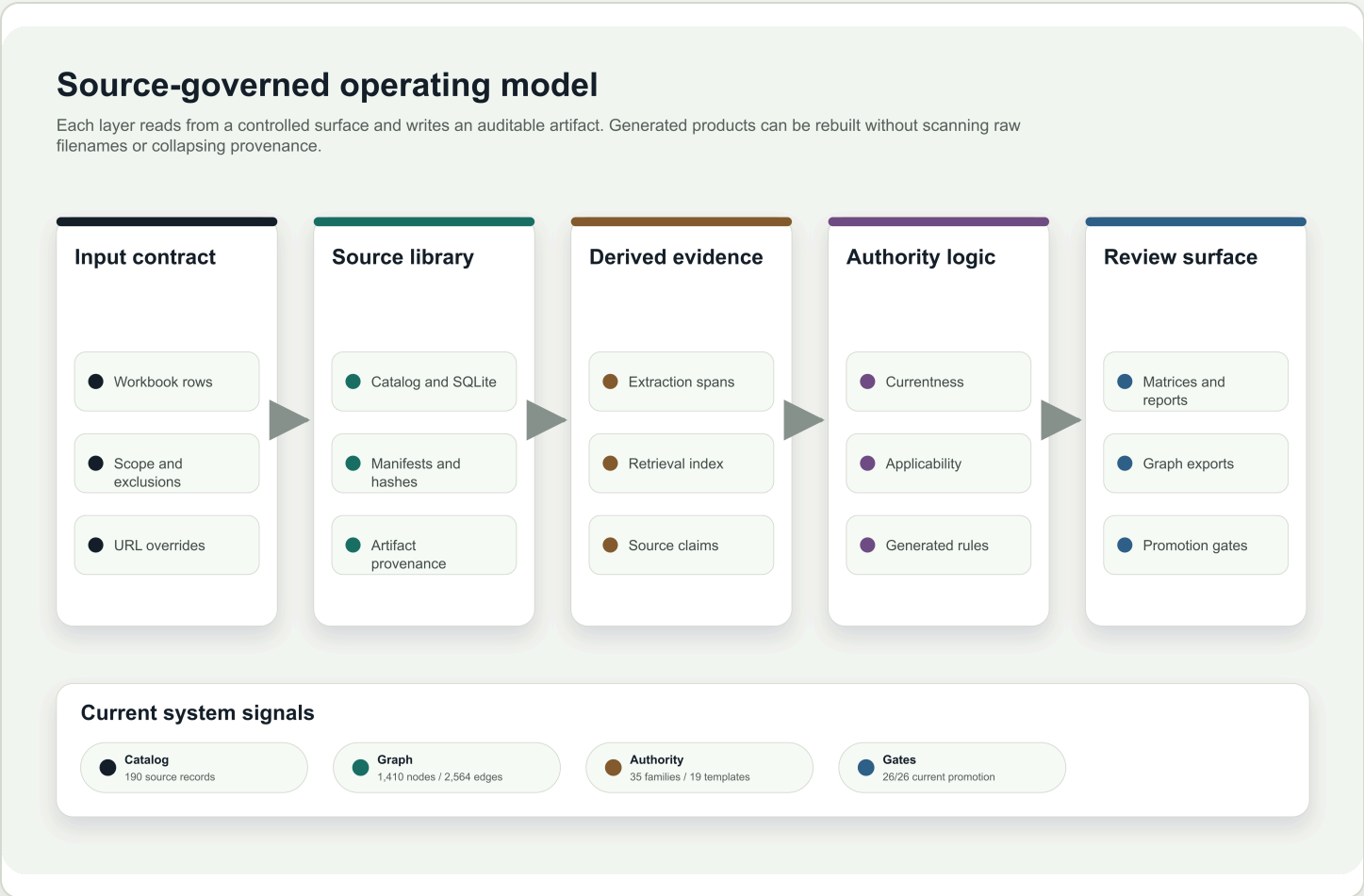
Architecture components at a glance

A source-governed review engine with traceable evidence, explicit authority logic, and gated outputs.



Controlled source library to graph-ready review layer

The workbook remains the contract. Downloader, catalog, extraction, retrieval, evidence graph, source-claim graph, rule binding, and knowledge graph exports all preserve row identity and provenance so downstream review products can be replayed and inspected.



Operating model: source records are captured once, then promoted through derived layers with explicit validation. The graphic separates source capture, semantic extraction, applicability, review products, and visualization so the same evidence model supports multiple outputs without duplicating logic.

Currentness and partitioning

Active review sources are separated from candidate, blocked, superseded, and currentness-only records before graph export. That prevents stale or excluded sources from silently supporting current authority.

Validation before display

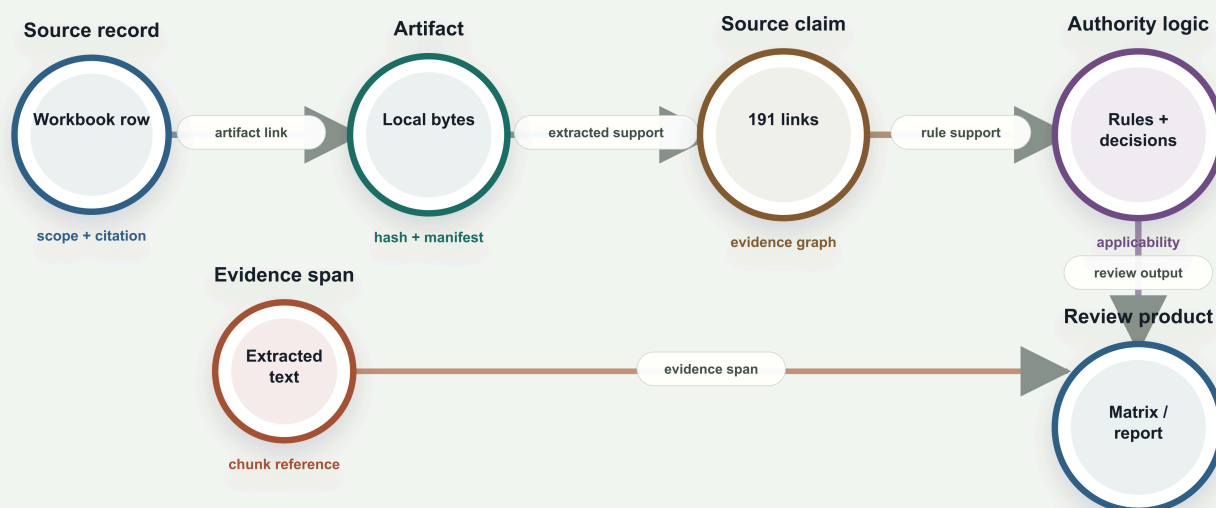
Graph exports carry node and edge type checks, provenance requirements, endpoint compatibility checks, readiness status, and failure-category counts before they are used by the viewer or promotion suite.

Every finding stays connected to evidence

The system keeps source records, artifacts, chunks, evidence spans, source claims, rule templates, applicability decisions, generated rules, and findings as graph-visible objects. Review outputs are produced from those artifacts rather than from a narrative-only analysis pass.

Evidence path stays inspectable

The system treats a finding as the end of a traceable path, not as a standalone sentence. Each link has a provenance role and can be tested.



Review outputs from the trace

- compliance matrix
- decision-support report
- final QA packet
- knowledge graph export
- 3D viewer scenes

Traceability model: source evidence supports claims; claims support rules and applicability; applicability drives review products. The chain remains inspectable across JSON, Markdown, PDF, SQLite, and graph export surfaces.

Reviewer-facing outputs

- Compliance matrix in JSON, Markdown, and PDF.
- Decision-support report with evidence paths and residual risk.
- Final QA certification and replay validation artifacts.
- Knowledge graph JSON, node/edge files, summaries, and validation.

Reverse compliance

The same evidence model can search for unsupported claims, older-regulation dependencies, missing source support, unresolved applicability, and forest-plan component gaps before the package is treated as reviewer-ready.

Promotion gates make readiness explicit

Readiness is not inferred from a finished-looking report. The system uses deterministic gates for source capture, currentness, extraction, retrieval, graph validation, applicability, compliance outputs, forest-plan evaluation, decision support, final QA, and promotion.

Readiness is a gated state

Reports, PDFs, and viewer scenes are downstream products. The system first verifies source capture, evidence, graph validity, review outputs, and promotion status.

Current gate stack

Catalog integrity
Workbook coverage, source statuses, artifact links

PASS

Currentness
Active, candidate, superseded, and blocked source roles

PASS

Graph validation
62/62 source-set checks

PASS

Phase evaluation
20/20 review phases

PASS

Promotion suite
26/26 current gates

PASS

Expansion boundary

Forest-plan profiles

1/10

graph-ready

Blocked profiles

9

visible readiness blockers

Components

329

inventory-backed

Profile requirements

8

directives + overlays

Blocked or incomplete sources remain explicit so expansion risk is visible before a package is called ready.

Source control

Workbook rows, URL overrides, catalog records, hashes, and source-set manifests.

Authority model

Authority families, currentness decisions, rule templates, and source partitions.

Review outputs

Compliance matrix, decision support, final QA, and reviewer-facing PDFs.

Evidence build

Extraction, retrieval, evidence spans, graph edges, and source-claim links.

Applicability

Package-specific decisions, generated rules, screened-out authority, and unresolved items.

Viewer surface

3D graph scenes driven by validated graph JSON rather than hard-coded visuals.

Expansion boundary

New packages, forest profiles, or authority deltas are added by expanding the source/profile contract and replaying gates. Blockers remain visible when profiles or sources are not ready, instead of being collapsed into a generic failure.

Boundary statement: the system produces auditable review support and readiness evidence. It does not replace professional judgment, agency review, or legal sufficiency review.